MAKEUP EXAM 2025

Algebraic Structures

1A. Let G be a group. Solve simultaneously, $x^2 = b$ and $x^5 = e$ for all x, b, e $\in$ G.

1B. Let G be the additive group of real numbers. Verify whether the following are the subgroups of G. (1) H = {log a: a $\in$ Q and a >0}. (2) K = {x $\in$ R: tan x $\in$ Q}.

1C. Find all zero-divisors and units in the ring $(Z_{36}, +, \cdot)$

2A. Prove that a finite integral domain is a field. Give an example of an integral domain which is not a field.

2B. Find all generators and all subgroups of additive group Z_{36}.

2C. Let G = R $\times$ R. Describe the quotient group with respect to H = {(x, y): y = -x}, the subgroup of G.

3A. Let f: R $\times$ R to $M_2(R)$ defined by f(x, y) = $\begin{pmatrix} x & 0 \\ 0 & y \end{pmatrix}$. Verify whether f is a homomorphism. If so, find its kernal.

3B. Let A = $Z_2 \times Z_6$ and J = {(0, 0), (0, 2), (0, 4)}. Compute the quotient group A/J.

3C. Prove a subgroup H of a group G is normal if and only if product of two left cosets is again a left coset.

4A. Show that $f(x) = 4x + 2$ is un-factorable in $Z_7[x]$. Find all solutions of $x^2 + 11x - 60 = 0$ in the rings $(Z_{15}, +, \cdot)$ and $(Z_{11}, +, \cdot)$ if exists.

4B. Determine if each polynomial is factorable or un-factorable in the specified rings.

1. $x^3 + x^2 + x + 1$ in $R[x]$.
2. $x^3 + 2x + 1$ in $Z_3[x]$.

4C. Prove that a ring has cancellable property if and only if it has no zero divisors of zero. Whether the rings Z_{12} and Z_{13} are integral domains.

5A. Express each of the following permutations in S_9 and S_8 respectively as a product of transpositions.

1. $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 4 & 9 & 2 & 5 & 1 & 7 & 6 & 8 & 3 \end{pmatrix}$
2. $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 8 & 2 & 6 & 5 & 1 & 7 & 4 \end{pmatrix}$

5B. Let G be a commutative group $\{e, a, b, b^2, ab, ab^2\}$ whose generators satisfy: $a^2 = b^3 = e$, $ba = ab^2$. Write the table for G.

5C. Let R be the set of real numbers. Define a binary operation * on R^+ as $x * y = \frac{xy}{x+y+1}$. Verify whether * is commutative, associative ? Whether identity exists ?